

SALMI, KuwaitWMO: **405700**Lat: **29.100N**Lon: **46.683E**Elev: **290**StdP: **97.90**Time Zone: **3.00 (E03)**Period: **82-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.2	4.3	-13.7	1.2	10.8	-11.8	1.4	12.3	12.0	12.9	10.7	15.7	4.0	300	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	47.0	20.6	46.1	20.2	45.0	19.9	22.5	37.1	21.4	40.1	20.8	41.0	5.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.3	27.2	17.0	12.6	25.4	15.5	11.4	22.9	67.7	35.7	63.5	40.1	61.2	41.2	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.7	9.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	-3.8	25.5	2.8	1.3	-5.8	26.4	-7.4	27.1	-8.9	27.9	-10.9	28.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	12.9	15.4	20.2	25.3	31.7	35.4	36.9	36.9	33.1	28.0	19.2	14.2
	DBStd	9.41	4.05	3.92	4.36	3.48	3.19	3.06	2.86	2.71	3.06	3.18	3.96	3.20
	HDD10.0	29	18	6	0	0	0	0	0	0	0	0	1	5
	HDD18.3	466	171	97	29	1	0	0	0	0	0	0	37	131
	CDD10.0	5807	109	156	315	459	674	763	835	835	693	557	275	136
	CDD18.3	3202	3	13	86	210	416	513	577	577	443	298	61	4
	CDH23.3	51212	52	124	933	2467	6462	8953	10381	10327	7168	3736	561	49
	CDH26.7	35635	13	32	406	1240	4273	6620	7926	7871	4927	2128	194	6
Wind	WSAvg	4.4	3.9	4.4	4.6	4.7	4.6	5.2	5.5	4.9	4.0	3.8	3.7	3.8
Precipitation	PrecAvg	142	25	18	20	17	4	0	0	0	0	5	28	24
	PrecMax	254	65	54	82	64	21	3	4	1	1	25	159	82
	PrecMin	59	2	2	0	2	0	0	0	0	0	0	1	1
	PrecStd	51	17	15	19	13	5	0	1	0	0	7	35	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	30.4	37.3	39.2	44.7	47.1	48.1	47.8	46.1	40.5	33.4	27.3
		MCWB	15.5	14.6	15.5	17.6	19.2	20.4	20.8	20.9	20.7	18.5	15.7	14.2
	2%	DB	24.4	27.1	34.0	36.7	42.8	45.9	47.2	46.7	44.0	38.9	31.0	24.6
		MCWB	13.9	13.7	15.7	16.9	18.9	19.9	20.5	20.6	19.7	18.1	15.3	13.1
	5%	DB	21.9	24.3	31.1	34.7	41.1	44.8	46.2	45.8	42.3	37.3	29.0	22.3
		MCWB	12.8	12.7	15.0	16.3	18.4	19.7	20.2	20.3	19.0	17.6	14.5	12.8
	10%	DB	19.6	22.1	28.4	32.7	39.5	43.4	44.8	44.6	40.8	35.6	26.8	20.3
		MCWB	11.7	11.9	14.1	16.0	18.0	19.0	19.8	19.9	18.5	17.0	14.0	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.4	19.6	19.7	21.1	21.6	23.7	24.0	24.1	23.7	17.9	18.0
		MCDB	23.6	23.5	27.8	30.6	36.4	42.7	37.0	34.4	32.6	30.8	24.0	20.5
	2%	WB	15.5	16.0	17.9	18.7	20.1	20.6	21.7	22.2	21.5	21.1	16.7	16.5
		MCDB	21.7	21.4	26.5	29.9	35.5	44.0	42.2	39.4	39.6	28.0	24.5	20.6
	5%	WB	14.2	14.8	16.7	18.1	19.4	20.0	20.9	21.4	20.1	19.9	15.9	15.0
		MCDB	20.4	21.1	25.5	28.8	36.3	43.1	43.9	41.4	40.5	28.5	24.1	19.9
	10%	WB	12.5	13.2	15.5	17.4	18.7	19.4	20.3	20.8	19.0	18.8	15.1	13.7
		MCDB	18.7	20.3	25.9	28.2	36.6	42.3	43.2	42.0	39.6	30.6	24.2	18.4
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.8	12.9	12.6	13.2	13.9	14.0	14.2	14.6	13.8	11.9	11.5
		MCDBR	13.9	14.1	16.3	14.7	14.8	14.6	14.9	15.0	15.4	15.1	14.3	13.9
		MCWBR	6.7	6.0	6.4	5.3	5.1	5.6	5.4	5.5	5.9	5.8	6.1	6.8
		MCDBR	12.5	11.6	12.5	12.0	13.3	14.2	14.5	14.2	14.9	12.8	11.5	11.1
	5% WB	MCWBR	6.7	5.9	5.4	4.4	5.0	5.8	6.0	5.7	6.3	5.3	5.5	6.0
Clear-Sky Solar Irradiance	taub		0.473	0.523	0.584	0.676	0.689	0.636	0.635	0.595	0.552	0.555	0.526	0.479
	taud		1.804	1.699	1.577	1.462	1.450	1.512	1.535	1.606	1.680	1.675	1.724	1.802
	Ebn at Noon		726	729	717	671	666	699	698	720	733	690	667	693
	Edh at Noon		182	220	265	307	311	291	284	262	235	222	196	175
All-Sky Solar Radiation	RadAvg		3.60	4.48	5.57	6.12	6.94	7.79	7.50	7.10	6.32	5.01	3.72	3.30
	RadStd		0.24	0.25	0.23	0.27	0.31	0.26	0.20	0.15	0.19	0.22	0.29	0.28